

RecoMart Data Quality Report

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Database: C:\Users\neave\Documents\GitHub\recomart-recommender\data\recomart.db

Overall status: PASS WITH WARNINGS

Full-table SQL checks validate integrity across every stored row. Pandas supplies descriptive profiles and Great Expectations validates deterministic samples of up to 100,000 rows per dataset.

Dataset row counts

Dataset	Rows
bronze_category_tree	1,669
bronze_events	2,756,101
bronze_item_properties	20,275,902
feature_item_cooccurrence	357,790
feature_item_cooccurrence_versions	357,790
feature_item_pair_stats	1,904,167
feature_item_popularity	417,053
feature_item_popularity_versions	417,053
feature_registry	3
feature_user_activity	1,407,580
feature_user_activity_versions	1,407,580
gold_item_features	417,053
gold_user_item_features	2,145,179
silver_category_hierarchy	1,669
silver_products	417,053
silver_user_events	2,756,101

Full-table validation checks

Check	Severity	Status	Violations
schema_bronze_events	critical	PASS	0
schema_bronze_item_properties	critical	PASS	0
schema_bronze_category_tree	critical	PASS	0
schema_silver_user_events	critical	PASS	0
schema_silver_products	critical	PASS	0
schema_silver_category_hierarchy	critical	PASS	0
schema_gold_user_item_features	critical	PASS	0
schema_gold_item_features	critical	PASS	0
missing_bronze_event_keys	critical	PASS	0
duplicate_bronze_events	warning	FAIL	460
invalid_silver_event_types	critical	PASS	0
missing_silver_event_keys	critical	PASS	0
transaction_id_consistency	critical	PASS	0
availability_domain	critical	PASS	0
duplicate_silver_products	critical	PASS	0
orphan_event_items	warning	FAIL	255,585
orphan_parent_categories	warning	PASS	0
missing_property_values	critical	PASS	0
duplicate_gold_user_items	critical	PASS	0
interaction_score_formula	critical	PASS	0
missing_gold_vectors	critical	PASS	0
bronze_to_silver_rejections	information	PASS	0

Pandas profiling summary

Dataset	Full rows	Sample	Total null cells	Duplicates
silver_user_events	2,756,101	100,000	99,347	17
silver_products	417,053	100,000	0	0
gold_user_item_features	2,145,179	100,000	0	0
gold_item_features	417,053	100,000	50	0

Null-count interpretation

Total null cells is a descriptive profiling count, not a count of validation failures. In the Silver event sample, 99,347 null cells are in transaction_id. Transaction events are a small minority of all interactions, and transaction_id is intentionally null for view and addtocart events. The full-table transaction_id_consistency check separately verifies that transactions have an ID and non-transaction events do not.

The Gold item sample also contains 50 null parent_category_id values, which are valid for root categories with no parent. Required identifiers, timestamps and model vectors are validated separately; unexpected nulls in those fields fail their corresponding checks.

Great Expectations results

Dataset	Expectation	Column	Status	Unexpected
silver_user_events	ExpectColumnValuesToNotBeNull	visitor_id	PASS	0
silver_user_events	ExpectColumnValuesToNotBeNull	item_id	PASS	0
silver_user_events	ExpectColumnValuesToNotBeNull	event_timestamp_ms	PASS	0
silver_user_events	ExpectColumnValuesToBeInSet	event_type	PASS	0
silver_products	ExpectColumnValuesToNotBeNull	item_id	PASS	0
silver_products	ExpectColumnValuesToBeUnique	item_id	PASS	0
silver_products	ExpectColumnValuesToBeInSet	available	PASS	0
gold_user_item_features	ExpectColumnValuesToNotBeNull	visitor_id	PASS	0
gold_user_item_features	ExpectColumnValuesToNotBeNull	item_id	PASS	0
gold_user_item_features	ExpectColumnValuesToBeBetween	interaction_score	PASS	0
gold_item_features	ExpectColumnValuesToBeUnique	item_id	PASS	0
gold_item_features	ExpectColumnValuesToNotBeNull	item_feature_vector	PASS	0

Issues and interpretation

WARNING: duplicate_bronze_events - 460 violations. Exact duplicate Bronze events are reported as a warning.

WARNING: orphan_event_items - 255,585 violations. Silver event items should exist in the product catalog.